

GYAN RANJAN

Full-Stack Developer | AI/ML Engineer | Competitive Programmer

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EDUCATION

**Indian Institute of Engineering Science and Technology,
Shibpur**

B.Tech — Information Technology
Enrollment No: 2025ITB040

2025 – 2029
CGPA: 9.22 / 10

TECHNICAL SKILLS

Languages	Python, JavaScript, C++, HTML5, CSS3
AI / ML	PyTorch, Scikit-learn, FAISS, Qdrant, LangChain, Sentence Transformers, Hugging Face, RAG Pipelines, Gemini API, Groq (LLaMA 3.3 70B)
Web (MERN)	React.js, Node.js, Express.js, MongoDB, Mongoose, REST APIs, EJS, Passport.js
Tools & Infra	Git, GitHub, Docker, FastAPI, Flask, Render, Vercel, Cloudinary, Hugging Face Spaces
DSA / CP	Data Structures & Algorithms, Competitive Programming — Codeforces Pupil (1200+), 200+ problems solved

PROJECTS

GitHub RAG Agent

FastAPI · Qdrant · Gemini · Groq · LLaMA 3.3 70B

- Engineered a production-grade Retrieval-Augmented Generation (RAG) system that ingests any public GitHub repository and answers natural-language queries against its codebase with streamed responses.
- Implemented multi-query expansion using LLaMA 3.3 70B via Groq to improve retrieval recall across chunked repository content stored in Qdrant Cloud with Google Gemini embeddings.
- Deployed FastAPI backend on Render and static frontend on Vercel; pipeline handles embedding rate-limits, ChromaDB-to-Qdrant migration, and chunk-level metadata for precise attribution.

DocOnCall — AI Healthcare Assistance Platform

Node.js · Express · MongoDB · Gemini API · Groq

- Built a full-stack AI-powered healthcare platform enabling users to analyse symptoms, receive personalised medical guidance, and manage doctor consultations through an intuitive web interface.
- Integrated Google Gemini API and Groq SDK for real-time AI-driven symptom analysis, medical report summarisation, health tips, and a conversational chatbot with contextual awareness.
- Implemented secure user authentication via Passport.js, session management, and automated email notifications using Nodemailer; follows MVC architecture with RESTful routing.

Anime Recommender System

FastAPI · FAISS · Sentence Transformers · RapidFuzz · Docker

- Developed a content-based recommendation engine using Sentence Transformers for semantic embeddings and FAISS for sub-millisecond nearest-neighbour search across a large anime catalogue.

- Integrated RapidFuzz for fuzzy title matching to handle misspelt queries; containerised with Docker and deployed on Hugging Face Spaces for public access.

Plant Disease Classifier

PyTorch · ResNet18 · FastAPI · Docker · Hugging Face

- Fine-tuned a ResNet18 convolutional neural network on a multi-class plant disease dataset using PyTorch, achieving high classification accuracy across 38 disease categories.
- Wrapped the model in a FastAPI inference server, containerised with Docker, and deployed on Hugging Face Spaces with an interactive upload-and-predict interface.

Bengaluru House Price Predictor

Random Forest · Flask · Scikit-learn · Render

- Trained a Random Forest regression model on Bengaluru real estate data with feature engineering (outlier removal, location encoding, BHK normalisation) to predict property prices accurately.
- Built a Flask REST API exposing the model, deployed on Render with a responsive frontend for real-time price estimation based on area, location, and room configuration.

SweetSuite — Property Rental & Listing Platform

Node.js · Express · MongoDB · Cloudinary · Passport.js

- Engineered a full-stack property rental platform supporting end-to-end listing management, image uploads via Cloudinary, user reviews, and role-based access control.
- Implemented secure authentication using Passport.js with Express Sessions; designed MVC architecture with RESTful routing, Joi validation, and server-side rendering via EJS.

ACHIEVEMENTS & CERTIFICATIONS

- 🏆 Best First-Year Team — HackSprint Hackathon, IEST Shibpur (Nov 2025) — awarded ₹5,000 prize along with certificate of recognition.
- Ignite Bootcamp: Idea to Plan — Wadhvani Foundation (Jun 2026) — completed 18-hour intensive entrepreneurship training covering ideation, prototyping, business modelling, and financial planning.
- Certificate of Participation — Zomathon by Zomato (Mar 2026).
- RootAccess CTF 2026 — Revelation 5.0, Unstop (Feb 2026) — participated as part of a team in the Braindead AI ML Hackathon.
- HackSprint Hackathon participation certificate (Nov 2025).

EXTRACURRICULAR ACTIVITIES

- Member, EDC (Entrepreneurship Development Cell), IEST Shibpur — actively engaged in startup ideation workshops, pitch events, and entrepreneurship-focused initiatives.
 - Member, CodeIEST (Coding Club), IEST Shibpur — participates in competitive programming contests, hackathons, and technical problem-solving sessions.
 - Member, DebSoc (Debating Society), IEST Shibpur — actively participates in debate competitions and public speaking events.
 - Participated in the college Model United Nations (MUN); represented Japan, deliberating on the policy of autonomous weapons systems.
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